

Statistical Models in Computational Biology

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Problem 29: Diploid genotype error model**(2 points)**

Consider the genotype error model introduced in Section 2.1 of the lecture, where the set of possible genotypes in a cell at a locus is $\mathcal{G} = \{AA, CC, GG, TT, AC, AG, AT, CG, CT, GT\}$, and ϵ is the per-site sequencing error, while δ is the per-site dropout rate. In the lecture, we calculated the probability $P(W = AA \mid Y = AG)$ of observing genotype AA given the true genotype AG .

Calculate the following probabilities analogously.

1. $W = TT$, and $Y = AG$.
2. $W = AT$, and $Y = AG$.
3. $W = AG$, and $Y = AG$.
4. $W = AA$, and $Y = AA$.
5. $W = AG$, and $Y = AA$.
6. $W = GG$, and $Y = AA$.

Problem 30: The Beta-Binomial distribution**(2 points)**

In the lecture, we learned about the Dirichlet-multinomial distribution $\text{DirMult}(X \mid n, \alpha')$ where α' is a vector of length ℓ . In the special case where $\ell = 2$, we write $\alpha' = (\alpha, \beta)$, and $\text{DirMult}(X \mid n, (\alpha, \beta))$ is also called the Beta-Binomial distribution, $\text{BetaBinom}(x \mid n, \alpha, \beta)$. It arises from the Binomial distribution for n Bernoulli trials in which the probability of heads is drawn from the beta distribution with shape parameters α and β .

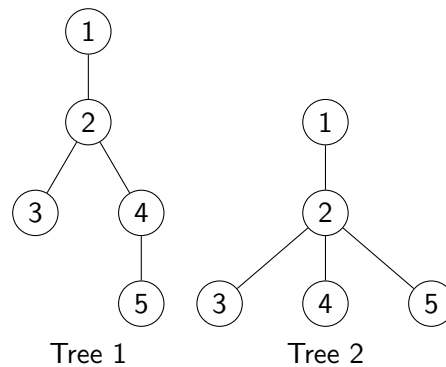
Plot a histogram for 2000 experiments drawn from each of the following distributions, with x axis being the number of successes for $n = 1000$ trials per experiment, and y axis being the number of experiments. Also write down: what is the mean and variance for each of these distributions?

1. Binomial distribution with head probability $p = 0.5$.
2. Beta-Binomial distribution with $\alpha = \beta = 1$.
3. Beta-Binomial distribution with $\alpha = \beta = 20$.
4. Beta-Binomial distribution with $\alpha = \beta = 1 * 10^6$.

Problem 31: Likelihood of binary genotypes given a tumor mutation tree (6 points)

The lecture claims that the likelihood of observed genotypes given a mutation tree marginalizing over all cell attachments can be computed in time $O(IJ)$, where I is the number of mutations and J is the number of cells.

1. Use the given R code skeleton, complete the implementation of a function that computes this log likelihood in time $O(IJ)$. (4 points)
2. Consider a toy example with 5 mutations and 7 cells. Load and inspect its genotype matrix from 'cells.txt'. Does this data permit a perfect phylogeny? Why? (1 point)
3. Once for false positive and negative rate $(\alpha, \beta) = (0.1, 0.01)$, and once for $(\alpha, \beta) = (0.01, 0.1)$, compute and report the log likelihood of this data given the two trees shown below, assuming uniform cell attachment probabilities. In each case, plot a node-by-cell heatmap for each tree, where the value is the likelihood of the observed genotype given the expected genotype of the node. (1 point)



4. Let us attach each cell to the node such that the likelihood of the observed genotype given the expected genotype is highest. Observe how different parameters cause the model to favor different tree structure and post-hoc cell attachment.